Assignment 4

10 pts



CSE Department
Computer Vision CMAI 480

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Assignment Instructions:

- The assignment must be turned electronically using Blackboard.
- This is an individual assignment -- each student should submit their own solution.
- In every uploaded file, include your name, date, and the assignment name.
- Use MATLAB for all tasks. Your script must run from beginning to end without manual repair.
- No makeups for the assignment are allowed.
- Please start early and ask me for help if you get stuck.
- **Authorized MATLAB Reference:** Students are encouraged to consult the MathWorks documentation example "Create Panorama" or https://www.mathworks.com/help/vision/ug/feature-based-panoramic-image-stitching.html and may use its explanations and MATLAB code as a reference while completing this assignment. The example demonstrates SIFT feature detection, feature matching, robust projective transformation estimation, transformation accumulation, center-image selection, and image warping. Students must adapt the workflow to their own images and satisfy all assignment requirements, including required visualizations, numerical results, bounded output dimensions, direct RGB pixel copying without blending, and independent answers to the conceptual questions.

Assignment 4: Image Stitching and Panoramic View Creation

Overview

In this assignment, you will extend feature-based image registration from a pair of images to a sequence of overlapping images. You will capture at least five images of the same scene, detect and match SIFT features in consecutive images, estimate projective transformations using RANSAC/MSAC, accumulate the transformations, register the color images in a common coordinate system, and directly combine the registered images into a panoramic view.

No blending is required. Visible seams, exposure differences, and abrupt changes in overlapping regions are acceptable and should be discussed. You will compare the manually generated panoramic view with a panorama captured using a mobile phone's built-in panorama mode.

Task 1: Capture the Input Images

Use a mobile camera to capture at least five overlapping images of the same scene.

- Capture the images in one direction, preferably from left to right.
- Maintain approximately 30% to 40% overlap between consecutive images.
- Rotate the camera slowly while keeping the camera near the same position. Avoid camera translation.
- Keep the camera approximately level. A tripod is helpful but not required.
- Choose a scene containing distinctive visual details such as corners, signs, textured surfaces, or irregular objects.
- Avoid large moving objects, very close foreground objects, large textureless areas, strong reflections, and rapid illumination changes.
- Save the files in capture order as image1.jpg, image2.jpg, image3.jpg, image4.jpg, and image5.jpg.
- Capture the same scene using the phone's built-in panorama mode and save it as MobilePanorama.jpg.

Required output: Display all captured images in their correct sequence.

Task 2: Load and Preprocess the Images

Load all images, resize them to a common size, retain the color images, and create grayscale copies for SIFT processing.

Use the following MATLAB code:

```
%% Task 2 - Load and Preprocess Images
clear; close all; clc;
rng(0);

imageFiles = {'image1.jpg','image2.jpg','image3.jpg', ...
              'image4.jpg','image5.jpg'};
numImages = numel(imageFiles);
assert(numImages >= 5,'At least five images are required.');

targetSize = [300 400]; % try smaller or bigger sizes [600 800] or other
colorImages = cell(numImages,1);
grayImages  = cell(numImages,1);

for n = 1:numImages
```

```
    assert(isfile(imageFiles{n}),'Missing file: %s',imageFiles{n});
    I = imread(imageFiles{n});
    I = imresize(I,targetSize);
    colorImages{n} = I;
    grayImages{n} = im2gray(I);
end

figure('Name','Input Image Sequence');
montage(colorImages);
title('Resized Input Images in Capture Order');
```

Question: Why must the color images be retained even though feature detection is performed on grayscale images?

Task 3: Detect and Match SIFT Features

Detect SIFT features in each image and match every image with the next image in the sequence.

Use the following initialization code:

```
%% Task 3 - Detect and Match Consecutive Images
previousGray = grayImages{1};
previousPoints = detectSIFTFeatures(previousGray);
[previousFeatures,previousValidPoints] = ...
    extractFeatures(previousGray,previousPoints);

fprintf('Image 1: %d SIFT features detected\n', ...
    previousPoints.Count);

tforms(numImages) = projtform2d;
tforms(1) = projtform2d(eye(3));
imageSize = zeros(numImages,2);
imageSize(1,:) = size(previousGray);
```

Suggested loop for consecutive image pairs:

```
for n = 2:numImages
    currentGray = grayImages{n};
    imageSize(n,:) = size(currentGray);

    currentPoints = detectSIFTFeatures(currentGray);
    [currentFeatures,currentValidPoints] = ...
        extractFeatures(currentGray,currentPoints);

    indexPairs = matchFeatures( ...
        currentFeatures,previousFeatures, ...
        MatchThreshold=40,MaxRatio=0.75,Unique=true);

    matchedCurrent = currentValidPoints(indexPairs(:,1));
    matchedPrevious = previousValidPoints(indexPairs(:,2));

    fprintf('Images %d-%d: %d putative matches\n', ...
        n-1,n,size(indexPairs,1));

    assert(size(indexPairs,1) >= 4, ...
        'Fewer than four matches between Images %d and %d.',n-1,n);

    figure('Name',sprintf('Matches %d-%d',n-1,n));
    showMatchedFeatures(colorImages{n},colorImages{n-1}, ...
        matchedCurrent,matchedPrevious,'montage');
    title(sprintf('Putative Matches: Images %d and %d',n-1,n));
```

Note: MaxRatio=0.75 is a starting value. A reasonable experimental range is 0.60 to 0.80. A smaller value is stricter and usually produces fewer matches.

Task 4: Estimate Projective Transformations Using RANSAC/MSAC

For each consecutive image pair, estimate a projective transformation that maps the current image to the previous image. Continue this code inside the loop from Task 3.

Use the following MATLAB code:

```
    [pairTform,inlierIdx,status] = estgeotform2d( ...
        matchedCurrent,matchedPrevious,'projective', ...
        MaxDistance=4,Confidence=99.9,MaxNumTrials=5000);

    assert(status == 0, ...
        'Transformation estimation failed for Images %d and %d.',n-1,n);

    tforms(n) = pairTform;

    numberOfMatches = size(indexPairs,1);
    numberOfInliers = nnz(inlierIdx);
    inlierRatio = numberOfInliers/numberOfMatches;

    fprintf('Images %d-%d: %d inliers (%.2f%%)\n', ...
        n-1,n,numberOfInliers,100*inlierRatio);

    figure('Name',sprintf('Inliers %d-%d',n-1,n));
    showMatchedFeatures(colorImages{n},colorImages{n-1}, ...
        matchedCurrent(inlierIdx),matchedPrevious(inlierIdx),'montage');
    title(sprintf('RANSAC/MSAC Inliers: Images %d and %d',n-1,n));

    % Accumulate the transformation into a common coordinate system
    tforms(n).A = tforms(n-1).A * tforms(n).A;

    % Prepare for the next image pair
    previousFeatures = currentFeatures;
    previousValidPoints = currentValidPoints;
end
```

Task 5: Select a Center Image

Select an image near the middle of the transformed sequence as the reference image. This reduces the overall distortion compared with using the first image as the reference.

You can use the following MATLAB code:

```
%% Task 5 - Select the Center Image
xlim = zeros(numImages,2);
ylim = zeros(numImages,2);

for n = 1:numImages
    [xlim(n,:),ylim(n,:)] = outputLimits(tforms(n), ...
        [1 imageSize(n,2)],[1 imageSize(n,1)]);
end

averageX = mean(xlim,2);
[~,sortOrder] = sort(averageX);
centerImageIdx = sortOrder(ceil(numImages/2));

fprintf('Center image: Image %d\n',centerImageIdx);
centerTformInv = invert(tforms(centerImageIdx));

for n = 1:numImages
    tforms(n).A = centerTformInv.A * tforms(n).A;
end
```

```
% Recalculate limits after changing the reference image
for n = 1:numImages
    [xlim(n,:),ylim(n,:)] = outputLimits(tforms(n), ...
        [1 imageSize(n,2)],[1 imageSize(n,1)]);
end
```

Question: Why is a middle image usually a better reference than the first image?

Task 6: Warp and Directly Combine the Images

Warp all resized color images into one common output coordinate system. Copy valid pixels directly into the combined image. No blending is required. If images overlap, pixels from the image processed later may replace earlier pixels. Visible seams are acceptable.

Suggested MATLAB code to define the common output reference:

```
%% Task 6 - Warp and Directly Combine Images
xMin = floor(min(xlim(:)));
xMax = ceil(max(xlim(:)));
yMin = floor(min(ylim(:)));
yMax = ceil(max(ylim(:)));

outputWidth  = xMax - xMin + 1;
outputHeight = yMax - yMin + 1;

outputRef = imref2d([outputHeight outputWidth], ...
    [xMin xMax],[yMin yMax]);

numberOfChannels = size(colorImages{1},3);
combinedImage = zeros(outputHeight,outputWidth, ...
    numberOfChannels,'like',colorImages{1});
combinedMask = false(outputHeight,outputWidth);
```

Suggested MATLAB code to warp and copy each image:

```
for n = 1:numImages
    I = colorImages{n};

    warpedImage = imwarp(I,tforms(n),OutputView=outputRef);

    inputMask = true(size(I,1),size(I,2));
    warpedMask = imwarp(inputMask,tforms(n),OutputView=outputRef);

    % Apply the same 2-D validity mask to every color channel
    for channel = 1:numberOfChannels
        outputChannel = combinedImage(:,:,channel);
        warpedChannel = warpedImage(:,:,channel);
        outputChannel(warpedMask) = warpedChannel(warpedMask);
        combinedImage(:,:,channel) = outputChannel;
    end

    combinedMask = combinedMask | warpedMask;
end

figure('Name','Manual Panoramic View');
imshow(combinedImage);
title('Directly Combined Panoramic View Without Blending');
imwrite(combinedImage,'ManualPanorama.jpg');
```

Important: Do not write combinedImage(warpedMask)=warpedImage(warpedMask) for an RGB image. The mask is two-dimensional and must be applied separately to the red, green, and blue channels, as shown above.

Task 7: Crop Large Empty Outer Borders

Use the combined validity mask to remove the large empty rectangular borders around the output. Irregular black regions may remain after projective warping, and that is acceptable.

Suggested MATLAB code:

```
%% Task 7 - Crop Large Empty Borders
validRows = find(any(combinedMask,2));
validColumns = find(any(combinedMask,1));

if ~isempty(validRows) && ~isempty(validColumns)
    croppedPanorama = combinedImage( ...
        validRows(1):validRows(end), ...
        validColumns(1):validColumns(end),:);
else
    croppedPanorama = combinedImage;
end

figure('Name','Cropped Manual Panorama');
imshow(croppedPanorama);
title('Cropped Panoramic View Without Blending');
imwrite(croppedPanorama,'ManualPanorama_Cropped.jpg');
```

Task 8: Compare With the Mobile-Phone Panorama

Load the panoramic view captured by the mobile phone and compare it with the manually generated result.

Suggested MATLAB code:

```
%% Task 8 - Compare With Mobile Panorama
assert(isfile('MobilePanorama.jpg'), ...
    'MobilePanorama.jpg was not found.');

mobilePanorama = imread('MobilePanorama.jpg');

figure('Name','Panorama Comparison');
subplot(2,1,1);
imshow(croppedPanorama);
title('Manually Created Panorama Without Blending');

subplot(2,1,2);
imshow(mobilePanorama);
title('Mobile-Phone Panorama');
```

In your comparison, discuss alignment, duplicated structures, geometric distortion, visible seams, brightness and color differences, cropping, field of view, and overall image quality.

Conceptual Questions

Add your answers as comments inside the MATLAB script.

Q1. Why should consecutive input images have approximately 30% to 40% overlap?

Q2. Why should the camera rotate around approximately the same position instead of translating between captures?

Q3. Why are SIFT features useful for matching overlapping images?

Q4. What is the purpose of the feature-matching ratio criterion?

Q5. What is the purpose of RANSAC/MSAC during projective transformation estimation?

Q6. Why must the pairwise transformations be accumulated when more than two images are registered?

Q7. Why is an image near the middle of the sequence selected as the reference image?

Q8. What may happen if two consecutive images have insufficient overlap?

Q9. Why can directly combined registered images contain visible seams?

Q10. What do duplicated edges or duplicated objects indicate about registration accuracy?

Q11. Does a visible seam necessarily mean that registration failed? Explain.

Q12. Why might the mobile-phone panorama appear more polished than the manually generated panorama?

Deliverables

Please do not compress or ZIP the files. Submit the following files individually:

- A completed MATLAB script named Assignment4_Fall2026.m, divided into executable sections beginning with %%.
- At least five ordered input images named image1.jpg through image5.jpg.
- The manually generated panoramic image named ManualPanorama.jpg.
- The cropped manually generated panoramic image named ManualPanorama_Cropped.jpg.
- The mobile-phone panoramic image named MobilePanorama.jpg.
- A PDF generated from the completed MATLAB script, figures, numerical results, and answers to all conceptual questions.

Submission Checklist

- Input images are ordered correctly and provide sufficient overlap.
- The script reports SIFT feature counts, putative match counts, inlier counts, and inlier ratios.
- Putative and inlier matches are displayed for consecutive image pairs.
- All transformations are accumulated and adjusted relative to a center image.
- The final panoramic view preserves RGB colors and uses direct pixel copying only.
- **No blending method is used.**
- The required output images are saved with the specified filenames.
- All conceptual answers are included as comments inside the MATLAB script.